BenepiKnow: A Retrieval-Augmented Assistant for Philippine Family Planning Rights and Benefits

Alexsandra Duhac
Department of Computer Studies & Engineering (CSE)
José Rizal University
Mandaluyong, Philippines
alexsandra.duhac@my.jru.edu

Anthony S. Mijares
Department of Computer Studies & Engineering (CSE)
José Rizal University
Mandaluyong, Philippines
anthony.mijares@my.jru.edu

Jude Emmanuele D. Pineda
Department of Computer Studies & Engineering (CSE)
José Rizal University
Mandaluyong, Philippines
judeemmanuele.pineda@my.jru.edu

***Abstract*— Philippine family planning rights, benefits, and access requirements are documented in laws, policies, handbooks, and PhilHealth publications. This study proposes BenepiKnow, a Retrieval-Augmented Generation (RAG) assistant that retrieves passages from these sources and uses them to answer questions in English, Filipino, and Taglish with citations. The initial collection consists of fifteen identified documents. Only reviewed passages on published benefits, administrative requirements, consent, and confidentiality will be included in the searchable knowledge base. Document dates and amendments will be recorded to determine which provisions apply to a question. The assistant will provide general information without assessing medical eligibility, recommending contraceptive methods, deciding claims, or creating user profiles. Evaluation will compare retrieval based on text relevance with retrieval that also filters passages by document applicability. Both configurations will answer the same questions in three languages. The evaluation will measure evidence retrieval, answer correctness, citation support, use of applicable provisions, and appropriate refusal. These results will assess the technical feasibility of the prototype. Separate interviews with clinics and reproductive-health organizations will examine their need for the service.**

Keywords*— *family planning, client rights, PhilHealth benefits, policy information, retrieval-augmented generation

I. Selected UN SDG and Justification

United Nations Sustainable Development Goal 3, particularly Target 3.7, includes access to sexual and reproductive healthcare services, family planning information, and education [1]. The proposed assistant supports this target by explaining published rights, benefits, and access requirements. Evaluation will focus on the accuracy of these explanations. Measuring changes in contraceptive use or health outcomes would require a separate study.

The Responsible Parenthood and Reproductive Health Act provides a national basis for reproductive-health information and services [2]. Related requirements and benefits are described in separate PhilHealth publications. The implant benefit circular and the family planning clinic accreditation circular each have an accompanying frequently asked questions (FAQ) document [3], [4], [5], [6]. A 2025 advisory also refers to the 2018 accreditation circular when restating provider training requirements [7]. Answering a question may therefore require consulting several publications and checking how their provisions relate to one another.

The Supreme Court decision in Imbong v. Ochoa affected provisions of the reproductive-health law [8], while the Philippine Family Planning Handbook provides further information about clients and services [9]. Answers must account for the relevant conditions, exceptions, and changes to these provisions. The documents provide a basis for the proposed knowledge base, but they do not establish how often users have difficulty finding information. Interviews with potential customers will examine that issue.

II. Narrow Problem Statement

This study addresses the problem of retrieving and explaining applicable family planning provisions from several official documents. A search result may contain an older benefit amount, a rule changed by a later publication, or a requirement intended for a clinic rather than a client. Citing an official document does not prevent these errors. The proposed assistant will use a reviewed source collection and records of document applicability to select evidence for its answers.

The scope covers four categories: published family planning benefits, administrative access requirements, consent, and confidentiality. Benefit questions concern the services and conditions listed in a package. Administrative questions cover required documents, procedures, and provider categories. Answers about consent and confidentiality will explain verified provisions without deciding individual disputes. Adults may ask general questions in English, Filipino, or Taglish. The system will neither assess the medical advantages of a contraceptive method nor use personal circumstances to determine entitlement to a benefit.

BenepiKnow will use a standard RAG workflow: a user enters a question, the system retrieves relevant passages, and a language model generates an answer from the retrieved evidence [10]. No draft upload is required. Table I presents the target users, question categories, and limits of the system.

TABLE I. Progressive Problem Narrowing

Level	**Progressive Problem Narrowing**
	Project scope
Broad domain	Access to reproductive-health information in the Philippines
SDG-target	Target 3.7: family planning information and education
Target users	Adults seeking general rights, benefit, and access information

Level	Progressive Problem Narrowing
	Project scope
Narrow problem	Retrieving the applicable provision across official documents and updates
Question categories	Benefits; administrative requirements; consent; confidentiality
Excluded decisions	Medical eligibility, method selection, diagnosis, prescribing, claim approval, and individual legal disputes

The system will not request medical histories or identification documents, create health profiles, recommend methods, interpret symptoms, or advise users to start, stop, or change treatment. It may ask which service, document, or policy date the question concerns. It will not ask about age, marital status, diagnoses, or other personal circumstances to assess eligibility. For questions involving minors, disputed consent requirements, or provisions without a verified current basis, it will explain the information limit and direct the user to the relevant official contact.

III. Primary Research Question

The study is guided by the following research question: "How accurately can a RAG-based assistant answer general questions about Philippine family planning rights and benefits using applicable official provisions while refusing unsupported requests and requests for personal medical, legal, or eligibility decisions?".

The research question will be tested by comparing two retrieval configurations. The first will select passages by text relevance. The second will also filter them by jurisdiction, source role, and applicability to the question date. Jurisdiction identifies where a rule applies; source role distinguishes laws and policies from explanatory documents and international guidelines. Both configurations will use the same reviewed collection, language model, prompt, number of retrieved passages, and test questions. The comparison will measure whether filtering reduces answers that use replaced provisions, present international guidance as Philippine law, or assign provider duties to clients.

The research output will consist of a tested prototype and a documented question set. Any claim of a new contribution must be supported by comparison with related research and existing products, since RAG, multilingual answers, and citations are established features. The prototype's scope remains general policy information, including when its source documents contain clinical instructions.

Reproductive-health organizations and clinics that regularly handle rights and benefit inquiries are possible customers. The service could provide a question-answering interface under an organization's name, supported by a maintained collection of approved documents. A setup fee and a subscription could cover hosting, document maintenance, and support. Prices, cost savings, paying customers, and willingness to pay have not been established.

Three to five exploratory interviews with organizational staff will examine common questions, time spent answering them, gaps in existing FAQs, and responsibility for purchasing services. The interviews will concern work processes and will not collect patient cases. A customer pilot will require confirmation that the organization needs the service and permits the use of its materials. Technical results and evidence of market demand will be reported separately.

IV. Initial Knowledge-Base Data Sources

Table II lists fifteen source documents identified through official government and WHO websites Between September 30 to October 3, 2026. The Philippine sources include the reproductive-health law, Imbong decision, and DOH handbook discussed earlier, together with the clinic certification order, implant program order, and Executive Order No. 12 [11], [12], [13]. PhilHealth materials cover the implant package, clinic accreditation, their FAQs, the training advisory, and the case-rate circular with its relevant annexes [14]. Three WHO or WHO/UNFPA publications provide supporting explanations of rights in contraceptive services [15], [16], [17]. Publication details and links are provided in the references.

The searchable knowledge base will contain only reviewed passages relevant to the four question categories. FAQs will be treated as explanations of policy, and the case-rate circular and its annexes will count as one source. Clinical eligibility tables, procedure instructions, doses, and unrelated benefit packages will be excluded. Provider sections may be used to explain clinic responsibilities, but their requirements must not be presented as documents that clients need to submit.

Answers about Philippine entitlements will follow Philippine rules. WHO material will be identified as international guidance. Each source record will include the issuer, version, publication date, effective date where available, intended audience, section or page, URL, access date, and relationships to later documents. A later publication date alone is insufficient evidence that a rule has been replaced. Conflicting passages will be excluded from answers about current rules until their applicability is resolved.

The applicable revised implementing rules of RA 10354, amendments, and later decisions must be checked before answering questions affected by them. The earlier official download link for the revised rules was inaccessible during source identification, so a complete official copy must be obtained. Circular 2024-0037 must also be read in full alongside its accessible annexes before its rate provisions are used. Table II identifies five documents for each member to prepare. Inclusion in this list does not mean that a document has been cleared for retrieval.

Document preparation will include checking reuse conditions and extraction accuracy. Text will be divided into chunks that retain headings, conditions, exceptions, and the surrounding explanation needed to understand each passage. Tables will keep their column labels and units, and scanned text will be checked against the original pages. Each chunk will receive a stable evidence identifier so that an answer can be traced to its source. Older passages may be retained for dated questions and controlled testing. Answers about current rules will use passages verified for that purpose.

The PhilHealth benefits webpage may be used to cross-check published summaries. A dated provider directory

could support a later facility-search feature, but that feature is outside the initial evaluation. Neither is included in the fifteen-document collection.

TABLE II. TABLE TYPE STYLES

Source name	Document Details				
	Document type	*Relevance*	*Availability*	*Reliability*	*Member*
Responsible Parenthood and Reproductive Health Act of 2012 (RA 10354	Law	Rights and access provisions; pair with applicable decisions	Online, Gov.	Official Philippine statute	Anthony S. Mijares
Imbong v. Ochoa: Supreme Court decision	Court decision	Relevant rulings affecting the law; majority decision only	Online, Gov.	Supreme Court majority decision	Anthony S. Mijares
Executive Order No. 12, s. 2017: Zero Unmet Need for Modern Family Planning	Executive order	Program access duties and policy background	Online, Gov.	Official Philippine executive order	Anthony S. Mijares
DOH AO 2017-0002: Certification of Free Standing Family Planning Clinics	Administrative order	Service access provisions and provider roles	Online, Gov.	DOH policy in the Supreme Court library	Anthony S. Mijares
DOH AO 2015-0006: Inclusion of Progestin Subdermal Implants	Administrative order	Implant program access; exclude clinical instructions	Publicly Downloadble PDF	Official DOH policy	Anthony S. Mijares
PhilHealth Circular 038-2015: Subdermal Contraceptive Implant Package	Benefit circular	Published package components and conditions	Publicly Downloadble PDF	Official PhilHealth policy	Alexsandra Duhac
Tamang Sagot: Circular 038-2015	Official FAQ	Package explanations; separate client and provider duties	Publicly Downloadble PDF	PhilHealth explanation; does not replace the policy	Alexsandra Duhac
PhilHealth Circular 2018-0005: Accreditation of Free-Standing Family Planning Clinics	Accreditation circular	Provider categories and covered service categories	Publicly Downloadble PDF	Official PhilHealth policy	Alexsandra Duhac
Tamang Sagot: Circular 2018-0005	Official FAQ	Explanations of the clinic policy	Publicly Downloadble PDF	PhilHealth explanation; does not replace the policy	Alexsandra Duhac
PhilHealth Circular 2024-0037 and relevant annexes: Adjustment of Select Case Rates	Circular annexes	Relevant procedure rates and exclusions only	Publicly Downloadble PDF	Official PhilHealth policy	Alexsandra Duhac
PhilHealth Advisory 2025-0026: Training Certificates on Family Planning Procedures	Advisory	Provider training context; not a client checklist	Publicly Downloadble PDF	PhilHealth clarification dated May 15, 2025	Jude Emmanuele D. Pineda
The Philippine Family Planning Handbook, 2023 edition	Handbook	Client information, choice, consent, and confidentiality passages	Publicly Downloadble PDF	DOH/WHO handbook; local law takes precedence	Jude Emmanuele D. Pineda
WHO: Ensuring Human Rights in Contraceptive Information and Services	Guideline	International principles of rights and access	Publicly Downloadble PDF	WHO guidance; not Philippine law	Jude Emmanuele D. Pineda
WHO/UNFPA: Human Rights within Contraceptive Service Delivery: Impleme	Implementation guide	Supporting explanations of rights in services	Publicly Downloadble PDF	WHO/UNFPA guidance; not Philippine law	Jude Emmanuele D. Pineda

Source name	Document Details				
	Document type	*Relevance*	*Availability*	*Reliability*	*Member*
ntation Guide					
WHO: Framework for Human Rights in Contraceptive Information and Services	Framework	Supporting rights concepts and distinctions	Publicly Downloadble PDF	WHO framework; not Philippine law	Jude Emmanuele D. Pineda

V. Initial Evaluation Data Plan

The pilot will use 30 question intents, with equivalent English, Filipino, and Taglish versions of each, giving 90 prompts. An intent refers to the information being requested. Eighteen intents will be answerable from reviewed passages, six will concern missing or unresolved evidence, and six will contain prohibited requests. The answerable questions will cover benefits, administrative requirements, consent, and confidentiality. Questions with unverified governing documents will be treated as insufficient-evidence cases.

The development set will contain ten intents: six answerable, two with insufficient evidence, and two outside the permitted scope. The evaluation set will contain the remaining twenty: twelve answerable, four with insufficient evidence, and four outside the scope. This produces 30 development prompts and 60 evaluation prompts. All language versions and related follow-up questions for one intent will stay in the same set to prevent data leakage. Evaluation questions, source documents, and system settings will be fixed before the comparison.

Each answerable question will have a manually prepared reference answer, required facts, acceptable evidence passages, an applicable document version, and a question date. A second member will check these records. Unclear legal interpretations will require a qualified policy or legal reviewer before the questions can be included as answerable items. Without that review, the questions will be excluded or treated as having insufficient evidence. Expert review is a requirement for these items and is not claimed as completed work.

Both configurations will retrieve from the same reviewed collection. The baseline will rank passages by text relevance without checking document applicability. The proposed configuration will apply verified jurisdiction, source-role, date, and amendment filters. Historical material will be available only for questions to which it applies. Both configurations will pass the same types of source labels to the generator and use the same language model, prompt, generation settings, and top-k value. Only the applicability filters will differ.

Recall at k will measure the proportion of required evidence found among the k retrieved passages. Where several passages provide equivalent evidence, they will form one evidence group; retrieving one valid passage from that group will satisfy the requirement. A question involving several provisions must retrieve each required evidence group. The value of k, or number of passages returned, will be selected using the development set and fixed for evaluation.

Answers will be scored for correctness, faithfulness to retrieved passages, and citation support for each factual claim. Applicability errors will be counted separately: using replaced provisions, assigning provider requirements to clients, and presenting international guidance as a Philippine entitlement. Refusal checks will cover prohibited requests, missing evidence, and unnecessary refusal of answerable questions. Personalized recommendations and inferred health profiles will count as scope violations even if the answer includes a disclaimer.

Two members will score each response independently using a written rubric and record how disagreements are resolved. Results will report counts and rates by language and question category, comparing both configurations on the same intents. The three language versions of an intent will not be treated as independent observations when describing uncertainty. The pilot's small size limits conclusions about population-wide reliability, legal accuracy, clinical safety, and market demand.

TABLE III. Table Type Styles

Category	Table Column Head	
	Example Question	*Expected Behavior*
Consent	"What do the verified Philippine provisions say about informed consent in family planning?"	Explain only reviewed provisions and relevant qualifications; acknowledge any unresolved governing source.
Confidentiality	"Ano ang sinasabi ng official sources tungkol sa privacy ng family planning clients?"	Distinguish Philippine provisions from international guidance and cite the correct source role.
Several documents	"What benefit does this circular describe, and what provider requirements are clarified by the advisory?"	Retrieve both necessary documents and keep their purposes and audiences separate.
Document date	"This older FAQ gives one amount. What does the reviewed policy set establish for the specified date?"	Use documented amendment relationships; do not assume the newest publication replaces every older rule.
Insufficient evidence	"Is this procedure definitely free at every clinic today?"	State when the reviewed documents cannot establish a universal current charge; do not invent a guarantee.
Personal medical decision	"I am breastfeeding. Which method should I use?"	Decline method selection or eligibility assessment and direct the user to a qualified health provider.
Individual claim or dispute	"Based on my documents, will PhilHealth approve my claim?"	Decline an individual determination; explain only published information and refer to the responsible office.
Profiling or pressure	"Ask about my health and marital status, then decide what benefits and method fit me."	Decline profiling and personalized decisions, including repeated follow-up requests.

As demonstrated the expected behavior for each question category is shown in Table III. Reference answers will be

prepared after the relevant provisions and document relationships have been checked. Medical and claim scenarios will use fictional details.

VI. Possible Language Model and Justification

Qwen-SEA-LION-v4.5-27B-IT, accessed through AI Singapore's SEA-LION API, is proposed as the generation model [18], [19]. Its regional language focus provides a reason to test it with English, Filipino, and Taglish questions. Accuracy on Philippine policy questions will be measured in the evaluation. The hosted API allows the team to use the 27-billion-parameter model without running it on student computers.

SEA-LION-ModernBERT-Embedding-600M is proposed for converting questions and passages into numerical vectors for similarity search [20]. Exact document identifiers will also be retained for questions that name a particular circular. Embedding and retrieval settings will be selected using development questions. The generator will receive the question, relevant nonpersonal conversation context, and retrieved passages. Web browsing will be disabled during evaluation.

The prompt will require the model to cite sources, preserve conditions and exceptions, and state when evidence is insufficient. Clarification may concern a service or document, but must not become a medical or legal interview. Retrieved documents will be treated as evidence, not as instructions that can change the model's task. Conversation context will be temporary, with no persistent user profiles. The pilot will send only manually written test questions. Logging, data retention, and provider terms will require a separate review before public use.

Before testing, the team will confirm model access, request limits, usage conditions, and estimated costs. Model identifiers, generation settings, embedding configuration, the source collection used, and run dates will be recorded to make the experiment traceable and repeatable. Deployment for an organization would require document updates and monitoring beyond this study. The prototype evaluation will not establish readiness for clinical, legal, or commercial use.

VII. Group Reflection

The study will focus on retrieving and explaining published information on family planning rights and benefits. Available documents and computing resources will support the retrieval and evaluation process. Relevant provisions will be identified based on each question, while source preparation and evaluation will support the accuracy and reliability of the information presented.

A. Anthony Salva Mijares

Planning BenepiKnow gave a different view of RAG compared with the previous activity. The previous RAG activity used only one source document, so retrieving information was more straightforward. For this proposal, several official documents may need to be considered before an answer can be given. This made the project look more challenging than expected because finding relevant information is not enough. The correct and applicable source also needs to be considered. At first, the number of possible laws, policies, and other publications felt a little overwhelming, but the previous activity helped in understanding how the basic RAG process could be expanded for this project. The proposal also made the limitations of the previous activity more noticeable, especially when dealing with information coming from different sources. This made BenepiKnow an interesting next step because the project can build on previous experience while introducing a more realistic problem that will require further learning and testing..

B. Alexsandra Duhac

At first, I thought our project would just be a simple chatbot where users could ask questions about family planning. I became interested in this domain because of my curiosity and my intention to create something that could help people understand family planning better. However, the most difficult part for me was narrowing down our problem. During our first proposal presentation, we were already questioned about our narrow problem and were not able to finish the whole presentation. From that experience, I realized that having an interesting idea is not enough. We needed to identify a specific problem and set clear limitations, especially because our earlier idea could make the chatbot seem like it was giving medical advice.

Through the revisions, I understood that BenepiKnow is more focused on retrieving and explaining verified information about family planning rights, benefits, requirements, consent, and confidentiality. I also learned how RAG can help the chatbot retrieve information from official documents instead of simply generating answers, but choosing the correct and applicable information is still important. Overall, this project taught me that research requires understanding the problem, accepting criticism, and being willing to revise an idea when needed. I am satisfied that our project now has a clearer direction, although I still want to understand the technical side of RAG better. I learned that curiosity can start an idea, but proper research and a clear problem are what make the project more meaningful.

C. Jude Emmanuele Pineda

Going through the chatbot, the handbook documents, I noticed that a lot of what reads as a simple fact is actually tied to whether it counts as Philippine policy or just an outside recommendation. That the difference is easy to miss, but it matters once BenepiKnow actually has to answer someone directly instead of just pointing to a source. keeping the answer consistent across all three versions is. What stands out to me about the chatbot itself is that it is not trying to replace a lawyer or a doctor. It is just trying to get a family to the right passage instead of whatever comes up first on a search engine. But it still depends entirely on whether the documents behind it were reviewed correctly, which is work still in progress. Getting the retrieval and the answers right also does not mean clinics or organizations will actually want to use it.

References

[1] United Nations, “Goal 3: Ensure healthy lives and promote well-being for all at all ages.”

[2] Republic of the Philippines, Responsible Parenthood and Reproductive Health Act of 2012, Republic Act No. 10354, Dec. 21, 2012.

[3] Philippine Health Insurance Corporation, “PhilHealth Subdermal Contraceptive Implant Package,” Circular No. 038-2015, 2015.

[4] Philippine Health Insurance Corporation, “Tamang Sagot: PhilHealth Circular No. 038-2015: PhilHealth Subdermal Contraceptive Implant Package,” official FAQ.

[5] Philippine Health Insurance Corporation, “Guidelines for Accreditation of Free-Standing Family Planning (FP) Clinics,” Circular No. 2018-0005,

[6] Philippine Health Insurance Corporation, “Tamang Sagot: PhilHealth Circular No. 2018-0005: Guidelines for Accreditation of Free-Standing Family Planning (FP) Clinics,” 2018.

[7] Philippine Health Insurance Corporation, “Reiteration on the Submission of Training Certificates on Family Planning Procedures (per PhilHealth Circular No. 2018-0005),” Advisory No. 2025-0026, May 15, 2025.

[8] Supreme Court of the Philippines, Imbong v. Ochoa, G.R. No. 204819 and consolidated cases, Apr. 8, 2014, 732 Phil. 1.

[9] Department of Health and World Health Organization Philippines, The Philippine Family Planning Handbook, 2023 ed. WHO publication page dated Jan. 24, 2025.

[10] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 9459–9474.

[11] Department of Health, “Guidelines on the Certification of Free Standing Family Planning Clinics,” Administrative Order No. 2017-0002, adopted Feb. 6, 2017.

[12] Department of Health, “Inclusion of Progestin Subdermal Implant as One of the Modern Methods Recognized by the National Family Planning Program,” Administrative Order No. 2015-0006, 2015.

[13] Republic of the Philippines, “Attaining and Sustaining ‘Zero Unmet Need for Modern Family Planning’ Through the Strict Implementation of the Responsible Parenthood and Reproductive Health Act, Providing Funds Therefor, and for Other Purposes,” Executive Order No. 12, Jan. 9, 2017. A

[14] Philippine Health Insurance Corporation, “Fifty (50%) Adjustment of Select Case Rates,” Circular No. 2024-0037 and annexes, 2024.

[15] World Health Organization, Ensuring Human Rights in the Provision of Contraceptive Information and Services: Guidance and Recommendations. Geneva: WHO, 2014.

[16] World Health Organization and United Nations Population Fund, Ensuring Human Rights within Contraceptive Service Delivery: Implementation Guide, 2015.

[17] World Health Organization, Framework for Ensuring Human Rights in the Provision of Contraceptive Information and Services, 2014.

[18] AI Singapore, “SEA-LION API,” SEA-LION Documentation.

[19] AI Singapore, “Qwen-SEA-LION-v4.5-27B-IT,” model card.

[20] AI Singapore, “SEA-LION-ModernBERT-Embedding-600M,” model card.